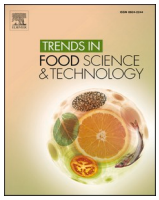
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Revolutionizing food safety in the airline industry: AI-powered smart solutions

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ABSTRACT

Background: Ensuring food safety in the airline industry is a complex challenge due to the global supply chain, long-distance transport, and the need for real-time monitoring. Traditional practices often fail to address inefficiencies, leading to foodborne illnesses, waste, and regulatory breaches. AI offers revolutionary potential by addressing these issues with cutting-edge technologies like IoT sensors, blockchain, and predictive analytics.

Scope and approach: This review explores the integration of AI in enhancing airline food safety. It examines AI-driven solutions across supply chain management, real-time monitoring, and predictive maintenance. Specific applications like automating compliance, mitigating risks in transport, and enhancing transparency through traceability are discussed. The paper highlights the role of AI in reducing food waste by up to 30 % through precise demand forecasting and controlling losses attributed to transport inefficiencies, which globally cause 20 % of food spoilage.

Key findings and conclusions: AI enhances operational efficiency by enabling real-time risk detection, optimizing inventory, and ensuring compliance with international safety standards. For instance, machine learning and IoT sensors significantly improve traceability and reduce spoilage costs, potentially lowering operational expenses by 15–20 %. Despite initial challenges like implementation costs and data security concerns, phased integration and cross-industry collaboration can overcome barriers. This paper concludes that AI-driven solutions are indispensable for ensuring long-term sustainability, reducing food waste, and meeting evolving regulatory and consumer expectations in airline food safety.

1. Introduction

Food safety in the airline industry is paramount (Azkat, 2024), given the unique challenges of preparing, storing, transporting, and serving meals at high altitudes under strict time constraints and regulatory scrutiny (Rajaratnam, 2023). Ensuring food safety is not only a legal and ethical obligation for airlines but also crucial for passenger health and satisfaction. Contaminated food can lead to health risks ranging from mild foodborne illnesses (Kanwal et al., 2024) to severe health crises, particularly as flights often transport diverse groups of individuals, some of whom may have vulnerable immune systems or dietary restrictions. Airline food safety is complex due to early preparation, multiple handovers, and varied storage conditions (Al Balushi, 2020; Asim & Yasmeen, 2021). Food handling, temperature control, cross-contamination, and allergen management are just a few of the factors (Brinkmann & Klug, 2020) that must be carefully monitored to prevent spoilage or contamination (Nantob-Bikatui, 2020). Additionally, airline meals are often produced in large quantities and stored for extended periods, which requires stringent temperature controls and protocols to minimize bacterial growth and ensure freshness. One of the primary hurdles to achieving consistently safe food on airlines is the difficulty of real-time monitoring and controlling safety parameters once food leaves the ground. Since food is prepared on the ground hours before it is consumed, it is exposed to varied temperature zones, from airport kitchens (van der Walt & Bean, 2022) to airplane galleys, where refrigeration options are often limited. The quality and safety of food can

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degrade during transportation, storage, or in-flight service, heightening the risk of foodborne pathogens. Moreover, strict regulations vary by country, further complicating food safety practices, as airlines must comply with multiple national standards. Additionally, the confined and unique environment of an aircraft makes it difficult to implement standard food safety procedures that are often feasible in more stable, on-ground facilities. Amid these challenges, Artificial Intelligence (AI) is emerging as a powerful tool to help airlines address food safety issues in new and innovative ways (L. Li, 2024). The integration of AI in food safety offers solutions that extend beyond traditional quality control and inspection practices by enhancing predictive capabilities, optimizing operational efficiencies, and providing real-time monitoring. AI-driven solutions such as machine learning, computer vision, and Internet of Things (IoT) devices allow airlines to improve their ability to track food from production to consumption, reducing risks associated with spoilage and contamination (Palakurti, 2022). Machine learning algorithms can analyze vast amounts of data from various stages of the food supply chain, identifying patterns and potential hazards that may go unnoticed by human oversight. For example, AI can analyze historical temperature and humidity data during food transportation and storage to predict when food is likely to spoil (Shi et al., 2024) or exceed safety parameters as presented in Fig. 1. This predictive capability allows airlines to preemptively address issues before they affect food quality or passenger safety. AI's role in food safety for airlines is also advancing with the use of IoT sensors that enable real-time temperature monitoring (Qian et al., 2023).

In recent years, significant progress has been made in integrating artificial intelligence (AI) technologies into the airline industry to enhance food safety (Abd Elmoaty Mohamed & Soliman, 2022). Several research studies have focused on the potential of AI applications such as machine learning, predictive analytics, Internet of Things (IoT) sensors, and blockchain to improve food monitoring, traceability, and real-time management of safety risks in the airline sector. These studies have demonstrated the usefulness of AI in ensuring food safety by automating processes, enhancing operational efficiency, and helping airlines comply with international food safety regulations. For instance, AI-powered systems have been employed for temperature and humidity monitoring during food storage and transport, ensuring compliance with safety standards. Predictive analytics has been explored for predicting food spoilage, enabling airlines to make data-driven decisions to minimize waste (Mortensen Ernits et al., 2022) and ensure food quality on board. Despite these advancements, research in this field still faces several gaps that need to be addressed. A major limitation is the lack of comprehensive AI integration across the entire food safety workflow in airlines, from preparation to in-flight management. While some components, like supply chain tracking and real-time monitoring, have been explored, few studies provide a holistic view of seamless AI implementation. Challenges such as setup costs, system compatibility, data quality inconsistencies and data privacy concerns are also underexplored. Airlines operate across multiple jurisdictions, each with its own food safety regulations, which add complexity to AI adoption. Additionally, most research focuses on AI's technological aspects, with limited attention to its practical applications, scalability, and the regulatory frameworks needed for successful adoption in the airline industry.

This review explores AI applications in airline food safety, examining both technological solutions and the operational and regulatory challenges of implementation. It provides an overview of the food safety workflow, from preparation and storage to supply chain management and in-flight monitoring. The paper highlights how AI technologies, including IoT, machine learning, and predictive maintenance, can

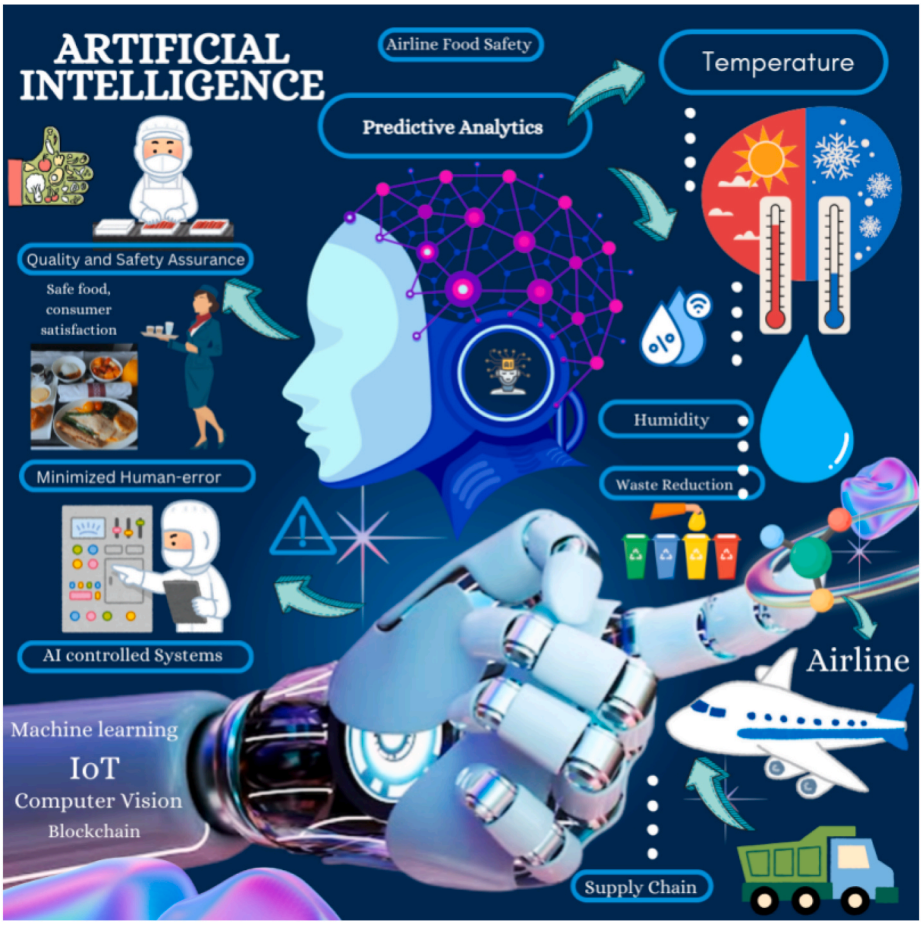


Fig. 1. Key Applications of AI in Airline Food safety.

enhance efficiency, safety, and sustainability in food handling. It also addresses the challenges airlines face in adopting AI, such as data privacy, integration costs, internet availability and regulatory compliance. While previous studies touch on these issues, this paper offers practical solutions and recommendations for successful implementation. The role of blockchain in food traceability and AI-driven predictive maintenance to prevent equipment failures are explored, areas underrepresented in existing literature. What distinguishes this review is its combination of technological insights with real-world challenges and solutions. It not only examines how AI can improve food safety but also addresses the operational, regulatory, and practical aspects of adoption. By highlighting emerging trends, this paper provides a forward-looking perspective on AI's impact on food safety in the airline industry, serving as a valuable resource for researchers, professionals, and policymakers.

2. Current food safety practices in airlines

In the airline industry, current food safety practices rely heavily on traditional protocols aimed at ensuring that food remains safe from preparation to consumption. These approaches follow Hazard Analysis and Critical Control Points (HACCP) principles, which systematically identify, assess, and manage food safety risks at critical control points like preparation, cooking, packaging, storage, and transportation (Rosak-Szyrocka & Abbase, 2020; Salesa et al., 2023). Airlines and their catering partners are required to maintain stringent hygiene practices, enforce safe cooking and storage temperatures, and follow strict timelines to ensure that food retains its quality and safety. Food handling staff undergoes regular training on hygiene (DAWOOD; Thamagasorn & Pharino, 2019) and safety practices, and physical inspections are conducted throughout the production and supply chain process to mitigate the risks of contamination, bacterial growth, or spoilage. Additionally, the food production and handling processes often adhere to international food safety standards, such as those set by the International Air Transport Association (IATA) (Поле‡кина, 2020) or regional regulatory authorities, which lay out requirements for food storage, transportation, and handling to avoid health hazards. However, these traditional practices face significant limitations when applied to the unique environment of air travel. One major challenge is the lack of real-time monitoring capabilities once food has left the ground and is on board the aircraft, making it difficult to detect and address potential safety hazards during flight. Unlike ground-based food service industries that can leverage continuous monitoring and quick response times, airlines are constrained by the logistical challenges of managing food safety under constantly changing temperatures and pressures during flight. Moreover, food is often prepared well in advance (Abdelhakim et al., 2019) and stored for extended periods before being served, which can result in the degradation of quality and increase the risk of contamination over time. These limitations in monitoring and response also mean that temperature fluctuations or deviations from safe handling procedures are harder to track, as onboard resources for temperature regulation and quality control are limited. Even with the best protocols in place, the industry's reliance on manual checks and periodic inspections leaves gaps in food safety, which can only be partially mitigated with current practices. As a result, the airline industry's reliance on traditional food safety protocols falls short of fully addressing the complexities of managing food safety in a high-altitude, closed environment with restricted control over variables like temperature and humidity. These constraints underscore the need for more advanced solutions, such as Artificial Intelligence (AI) (Yi Shien, 2022) and Internet of Things (IoT) applications, which could potentially provide real-time monitoring and predictive insights to improve food safety standards on flights. The industry's dependency on traditional methods, while generally effective for ground-based catering, highlights the limitations in its current food safety approach, especially in responding to in-flight risks dynamically and efficiently (Hussein et al., 2023). Table 1 presents food safety in Airlines with AI vs. without AI.

Table 1
Food Safety in Airlines: With AI vs. Without AI.

	Food Safety Aspects	With AI	Without AI
1	Contamination Detection	Real-time pathogen scanning & AI alerts (e.g., IBM's AI-assisted bacterial detection) (Qian et al., 2023).	Manual checks, slower contamination response
2	Temperature Monitoring	Automated real-time monitoring & adjustment (e.g., Walmart's IoT-based cold chain monitoring) (Krishnan et al., 2024).	Periodic manual checks, prone to inconsistency
3	Allergen Management	Precise allergen tracking & personalized alerts (e.g., Nima's AI-powered allergen sensor) (Adedeji et al., 2024).	Basic labeling, limited real-time allergen data
4	Inventory Control	AI-driven demand forecasting & optimized stock (e.g., Amazon Fresh using AI for inventory management).	Over/understocking issues, manual tracking
5	Food Freshness	Predictive freshness tracking based on AI analysis (e.g., smart expiration date labels used by Mimica) (X. Li et al., 2023).	Quality decline due to delayed spoilage detection
6	Regulatory Compliance	Automated compliance checks & record-keeping (e.g., Blockchain-based compliance in Nestlé's supply chain).	Risk of non-compliance, manual record handling
7	Waste Reduction	Predictive analysis for waste minimization (e.g., Too Good To Go app uses AI to connect surplus food with consumers) (Onyeaka et al., 2023).	High food waste due to inaccurate forecasting
8	Supply Chain Traceability	Real-time tracking from source to endpoint (e.g., IBM Food Trust blockchain tracing Walmart's leafy greens) (Dedeoglu et al., 2023).	Limited visibility, delayed response in recalls
9	Quality Assurance	AI quality control through image and sensor analysis (e.g., Alibaba's AI food inspection for detecting defects).	Labor-intensive quality checks, risk of oversight
10	Customer Safety Alerts	Instant notifications for recalls or health alerts (e.g., Carrefour's AI-driven product recall alerts) (Chhetri, 2024; Chhetri, 2024).	Delayed alert process, reduced response efficiency

3. AI-powered technologies in food safety

AI-powered technologies are rapidly transforming food safety practices in industries with complex supply chains, such as airlines, by providing innovative solutions that extend beyond the capabilities of traditional methods. Table 2 presents some examples. Among the most relevant AI applications in food safety are machine learning (D. Wang et al., 2023; X. Wang et al., 2022), the Internet of Things (IoT) (Bouzembrak et al., 2019; Mu et al., 2024), and predictive analytics. Machine learning allows systems to process and analyze vast amounts of data collected across the food production and distribution process, learning from historical patterns to identify potential risks before they arise. This capability is particularly useful in anticipating and preventing food safety issues by alerting staff when deviations in standard procedures, such as temperature fluctuations or increased humidity levels, could compromise food quality. IoT technology involves the use of connected sensors that monitor various environmental factors (Al Mamun & Yuce, 2019) in real-time, including temperature, humidity, and location, ensuring that food safety standards are maintained at all stages of the supply chain. These IoT-enabled sensors can relay data to

Table 2
AI-based technologies and their functions as food safety tools in the airline industry.

	Smart Technology	Function as a Food Safety Tool in Airlines
1	IoT Sensors	Real-time monitoring of temperature, humidity, and freshness to ensure food quality during storage and transport (Torres-Sánchez et al., 2020).
2	Predictive Analytics	Forecasts food demand, minimizes waste, and helps in inventory management by predicting stock needs.
3	Computer Vision	Automated inspection for contamination, food quality, and labeling accuracy, reducing human error (Hemamalini et al., 2022).
4	Blockchain	Provides secure, traceable records for the entire food supply chain, ensuring transparency and authenticity (Xu et al., 2022).
5	Machine Learning Models	Identifies patterns in food safety data to detect contamination risks and optimize storage conditions.
6	RFID Technology	Tracks and authenticates food packages throughout the supply chain, ensuring timely identification of issues.
7	Allergen Detection Kits	AI-enabled allergen identification in ingredients to prevent allergic reactions in passengers.
8	Automated Quality Control	Uses AI to scan and assess food quality standards, such as color, texture, and appearance, before serving.
9	Digital Twins	Simulates food supply chain scenarios to predict and mitigate risks of contamination or spoilage (Isuru et al., 2023).
10	Natural Language Processing (NLP)	Analyzes customer feedback on food safety, helping airlines improve quality and address safety concerns (Makridis et al., 2023).

centralized systems where AI algorithms assess whether food items remain within safe handling parameters, especially during storage and transportation phases, which are critical in the airline industry (Mu et al., 2024). Predictive analytics plays an essential role in forecasting potential safety issues by leveraging historical data and statistical models to predict when food items might begin to degrade. This proactive approach allows airline food service providers to address emerging problems before they affect passengers, minimizing the risk of foodborne illness or spoilage. The benefits of AI for food safety in the airline industry are substantial, particularly under the time-sensitive and logistical constraints (Whyte & Lohmann, 2020) inherent in air travel. Unlike traditional food safety methods, which are often reactive, AI-powered technologies are largely preventive and proactive, enabling airlines to maintain high safety standards even within challenging operational conditions. Real-time data analysis and alerts allow airlines to monitor food conditions continuously, taking immediate action to rectify issues rather than relying on manual inspections or scheduled checks. This approach helps airlines optimize the quality of food served at high altitudes, where resources and equipment are limited, and it minimizes the risk of safety incidents that could disrupt service or endanger passenger health. Furthermore, by reducing manual workload and automating monitoring processes, AI also enables airlines to allocate resources more efficiently, ultimately improving both the safety and efficiency of in-flight food service (Khudhair et al., 2019).

4. Applications of AI in the airline food safety workflow

AI applications within the airline food safety workflow are multifaceted, addressing various stages from food preparation and storage to real-time monitoring on flights and predictive maintenance. Together, these applications enable airlines to manage food safety more precisely and effectively, leveraging data-driven insights to ensure quality and compliance. Below is an expanded overview of these AI applications and their roles in the airline food safety workflow.

4.1. Food preparation and storage

AI plays a transformative role in food preparation and storage, enhancing the safety, consistency, and efficiency of these stages. In food preparation, AI-driven systems monitor cooking processes (Bidyalakshmi et al., 2024) to ensure that meals reach appropriate temperatures to eliminate potential pathogens, a key factor in preventing foodborne illnesses. Sensors integrated with AI systems (Q. Jiang et al., 2020; Qian et al., 2023) can regulate temperature and humidity levels in cooking and cooling environments, ensuring that the specific parameters required for safe food preparation are maintained throughout. This is particularly critical for airline catering, where food is prepared hours before consumption and stored until it can be transported. Additionally, AI can streamline food packaging (Trapani, Ma, & Jiao, 2023) by analyzing and automating aspects of the process to avoid cross-contamination, a prevalent issue in environments where large quantities of meals are processed at once. By integrating with IoT sensors, AI systems can also track the conditions of packaged food items, detecting if packaging integrity is compromised or if products deviate from safe storage temperatures. In terms of storage, IoT-enabled sensors connected to AI platforms are invaluable for continuously monitoring cold storage units, assessing real-time data on temperature and humidity to keep food at optimal freshness (Afreen & Bajwa, 2021). If a deviation occurs such as a refrigerator failure AI can trigger automated alerts, enabling immediate corrective action to prevent food spoilage. Furthermore, AI-driven data analysis can help catering managers identify patterns over time, such as areas where storage conditions are frequently compromised, allowing for proactive adjustments to equipment and protocols. This predictive oversight in both preparation and storage is essential in an industry where food safety is tied to extended storage times before consumption (Qian et al., 2023; Qu et al., 2022).

4.2. Supply chain management

AI-driven supply chain management for food safety in the airline industry is crucial in tracking and tracing ingredients from their source to the final in-flight meal as given in Fig. 2. Given the global nature of airline food supplies, ingredients come from various locations with unique handling and transportation conditions. AI technology monitors these processes, ensuring transparency and quality control throughout the supply chain. Machine learning algorithms analyze supplier data, flagging anomalies that could affect food safety (Chhetri, 2024). For example, if ingredients are exposed to improper temperatures during transport, AI systems can detect these issues before food preparation, reducing contamination or spoilage risks. Additionally, AI enhances food traceability by integrating with blockchain technology, allowing real-time tracking of each step from farming to distribution (Qin et al., 2023). This allows for precise tracking of each ingredient's origin and storage history, which is invaluable for addressing potential recalls or food safety investigations. Traceability also ensures compliance with international food safety regulations, as AI systems can instantly pull up records to verify that each product in the supply chain has met specific safety standards. In addition, AI's predictive analytics can help forecast inventory needs based on historical demand patterns (Nweje & Taiwo, 2025), reducing food wastage by minimizing overstock and ensuring only fresh ingredients are procured. This is particularly beneficial in the airline industry, where food supplies must be managed across a global network and forecasted accurately to meet the fluctuating demands of different routes and seasons. However, challenges such as data quality inconsistencies and the need for system interoperability across diverse international suppliers remain significant hurdles in AI adoption within global supply chains. For example, a recent case study in the airline industry demonstrated that AI-enabled traceability systems reduced food safety incidents by 30 %, but integrating data from various legacy systems posed initial difficulties (Vegesna et al., 2024).

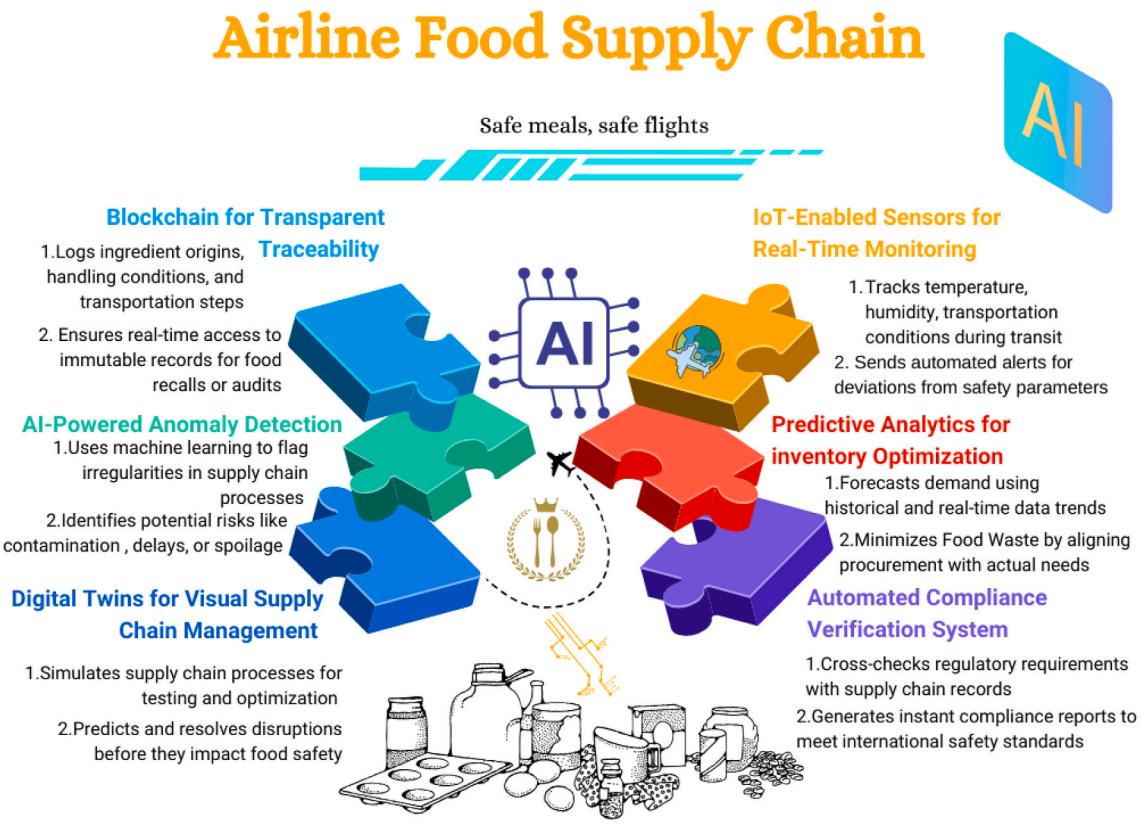


Fig. 2. AI in airline food safety within supply chain.

4.3. *Real-time monitoring on flights*

One of the most advanced applications of AI in airline food safety is its capability to conduct real-time monitoring during flights, a critical point where traditional monitoring approaches are limited due to inaccessibility and resource constraints. IoT sensors equipped with AI software are installed in onboard storage compartments and refrigerators to continuously monitor temperature and humidity levels, ensuring that food remains within safe conditions until it is served. These sensors communicate real-time data to a centralized system, allowing crew members to be immediately alerted if conditions deviate from safe thresholds. For example, if there is a malfunction in an onboard refrigerator (Raheja, Rathaur, & Faiz) or a storage unit door is left open, the AI-driven system can instantly notify the crew, allowing for prompt corrective actions to prevent food spoilage. This real-time monitoring capability is essential for perishable foods, as fluctuations in temperature can significantly impact safety and quality. Despite these advancements, IoT sensors and AI systems still face challenges in extreme or unpredictable flight conditions, such as sudden turbulence or electromagnetic interference, which can disrupt sensor accuracy and data transmission. Recent innovations, including more robust sensors designed to withstand harsh environments, are showing promise, but ongoing research is needed to enhance reliability under these conditions (Wei et al., 2024).

Additionally, AI systems can collect and analyze data across multiple flights to identify patterns or recurring issues in storage conditions (Misra et al., 2020), which can then be addressed through maintenance or procedural changes. This data-driven approach helps airlines maintain a high standard of food safety and reduces the chances of foodborne illness caused by improper storage. Beyond immediate food safety, real-time monitoring also allows for quality assurance, as the AI system can track any factors that might impact the taste or texture of food items (Obrist et al., 2019), ensuring that passengers receive meals that are both safe and enjoyable.

4.4. *Predictive maintenance*

Predictive maintenance, driven by AI, is another critical application that plays an essential role in safeguarding food quality by ensuring that equipment used in food preparation and storage operates reliably (Kisten et al., 2024). In the airline industry, catering facilities, storage units, and onboard refrigeration systems face constant strain due to frequent use and varying environmental conditions. Traditional maintenance schedules, based on fixed intervals, can be inefficient, as equipment may fail unexpectedly between services. AI-driven predictive maintenance shifts airlines from a reactive to a proactive model by forecasting potential failures. Machine learning algorithms analyze sensor data to detect early signs of wear or malfunction, such as temperature fluctuations, increased energy consumption, or unusual sounds. This allows airlines to schedule maintenance as needed, reducing the risk of failures that could spoil food. For instance, if a refrigeration unit shows signs of imminent failure, the AI system alerts ground staff to arrange repairs, ensuring meals on future flights remain safe (Kisten et al., 2024). By reducing the occurrence of unplanned equipment downtime, predictive maintenance contributes to consistent food safety standards (Uzoigwe, 2024), as airlines can avoid situations where food is stored in suboptimal conditions due to unforeseen equipment issues. Additionally, predictive maintenance reduces operational costs by extending the lifespan of equipment, as repairs and maintenance are conducted only when necessary, based on actual wear rather than arbitrary schedules. AI-driven predictive maintenance has proven to be cost-effective, with studies indicating a reduction in maintenance costs by up to 25 % due to fewer unplanned repairs and extended equipment lifespan (Ünlü & Söylemez, 2024). For example, airlines that implemented predictive maintenance for onboard refrigeration units saw a 20 % decrease in equipment failure rates (Dibsdale,

2020) and a 15 % reduction in maintenance costs over a three-year period.

In summary, AI applications in the airline food safety workflow covering food preparation and storage, supply chain management, real-time monitoring during flights, and predictive maintenance provide airlines with comprehensive and data-driven tools to ensure the safety, quality, and compliance of their in-flight meals (Ernits). By addressing each stage of the food safety workflow with precision and predictive capabilities, AI enhances food safety standards in the airline industry, providing solutions that are both preventative and responsive (Spanaki et al., 2022). As AI continues to evolve, its applications within the airline food safety workflow are likely to expand, further optimizing processes and setting new benchmarks for food safety in the industry.

5. AI for compliance and documentation

In the airline industry, compliance with food safety standards and effective documentation are essential to meet regulatory requirements and maintain passenger trust. The international nature of airline operations (Klophaus & Yu, 2023) means that food safety practices must adhere to stringent guidelines set by multiple regulatory bodies, such as the International Air Transport Association (IATA), the World Health Organization (WHO) (Azkat, 2024; Bicer, 2023), and national food safety authorities of the countries involved in each flight route. Achieving compliance with these standards requires rigorous record-keeping, traceability, and systematic documentation of food safety practices. This process, however, has traditionally been resource-intensive and prone to human error. Artificial intelligence (AI) has emerged as a powerful solution for automating and enhancing compliance management, providing airlines with the tools necessary to streamline record-keeping, ensure adherence to standards, and maintain transparency.

5.1. Automating record-keeping for regulatory compliance

AI plays a transformative role in automating record-keeping processes (Nakigozi, 2009), a fundamental aspect of food safety compliance in the airline industry. Traditionally, maintaining records related to food safety involved manual data entry and document management, requiring extensive labor and time. AI allows airlines to automate food safety processes, reducing personnel workload and minimizing errors (Wong & Man, 2023). Integrated with data capture tools, AI systems automatically log each step of food preparation, storage, and transportation, ensuring compliance with safety protocols. These systems track key variables like temperature, humidity, and handling practices, documenting the food's journey from preparation to consumption. For example, IoT sensors in storage and transportation units capture environmental data, which AI analyzes for compliance verification (Panduman et al., 2024). This data is automatically uploaded to a centralized system, creating a digital record (Yusriza, 2022) that can be reviewed by regulatory authorities or internal auditors.

In addition to real-time monitoring, AI can perform historical analysis, providing insights into trends and patterns in food safety practices over time. This capability enables airline operators to identify areas for improvement and proactively address compliance gaps, reducing the likelihood of violations or safety incidents. Moreover, AI-based systems offer automated audit trail generation, which is crucial in the event of a regulatory inspection or food safety incident. With automated records (Ndhlovu et al., 2024), airlines can promptly access relevant documentation to demonstrate compliance with regulatory requirements, improving the speed and accuracy of responses to regulatory inquiries. This is particularly valuable in the airline industry, where multiple international regulations may apply depending on the flight route. AI systems that integrate with blockchain technology provide additional benefits by enabling immutable record-keeping, ensuring that compliance records cannot be altered retroactively (Agrawal et al., 2025). This level of transparency and security in record-keeping builds confidence among regulatory bodies, passengers, and stakeholders.

5.2. Ensuring adherence to international food safety standards

In the highly regulated airline industry, adherence to international food safety standards is essential. Compliance requirements vary by jurisdiction, and airlines must ensure their practices meet the standards of all the countries they operate in. AI technology aids this by standardizing and automating compliance with diverse regulations, enabling airlines to efficiently manage complex international requirements (Klophaus & Yu, 2023). AI-driven systems help maintain consistent food safety practices aligned with the latest industry standards. These systems monitor conditions in real-time and send automated alerts if any process deviates from safety protocols, allowing for immediate corrective action. For example, food handling and preparation must meet hygiene standards set by organizations like the FDA in the U.S (Ebrahimian et al., 2022), the European Food Safety Authority (EFSA) in Europe (Cavalli et al., 2019), and the IATA guidelines for international flights. AI systems can cross-reference each stage of food preparation and storage with safety standards, ensuring all requirements are met. When deviations occur, AI generates automated corrective action plans to bring processes back into compliance, preventing minor issues from escalating (Muszyński et al., 2022). This automation helps airlines safeguard passenger health and stay compliant with regulations. AI also aids in updating practices in response to changing food safety regulations, a challenge for global airlines. AI can automatically integrate regulatory updates, ensuring compliance without manual intervention and reducing the risk of outdated practices. Additionally, AI supports predictive compliance by analyzing historical data to identify potential risks. For example, if a food product has a history of non-compliance, the AI system can flag it and suggest precautions (Reddy, Gupta, Kumar, & Kashyap). This predictive approach allows airlines to focus on high-risk areas, proactively addressing issues before violations occur.

In conclusion, by automating documentation processes, enabling real-time compliance monitoring, and supporting traceability, AI-driven systems provide airlines with the tools needed to navigate complex regulatory landscapes (Yilmaz, 2024) and maintain high standards of food safety. As the airline industry continues to evolve, AI's role in compliance and documentation will likely expand, offering new opportunities to optimize food safety practices and further strengthen industry standards.

6. Consumer safety and transparency

In the airline industry, consumer safety is paramount (Al-Shraideh, 2024; Islam et al., 2023), and transparency regarding food ingredients, allergens, and dietary information is increasingly expected by passengers worldwide. Airlines serve a vast and diverse customer base, many of whom have specific dietary needs, food allergies, or personal preferences. As such, providing clear and accurate information about in-flight meal components is essential to safeguarding passenger health (Grout & Speakman, 2021) and improving their overall experience. Traditionally, however, disseminating detailed food information has been challenging, especially under the unique conditions of air travel, where space, time, and resources are limited. Artificial intelligence (AI) has emerged as a transformative solution for improving consumer safety (Lamb et al., 2021) and transparency by facilitating real-time information sharing, automating ingredient labeling, and enhancing the airline's ability to respond to passenger dietary needs.

6.1. AI applications for labeling and information sharing about ingredients and allergens

AI offers powerful tools for automating and refining the labeling

process, especially in areas like ingredient transparency and allergen identification. By utilizing AI-powered systems, airlines can streamline the documentation of ingredient lists, food origins, and nutritional information for each menu item offered in-flight (Bugault, 2024). Through machine learning and natural language processing (NLP), AI can interpret and classify extensive ingredient lists (Hu et al., 2023), identifying potential allergens or dietary-restricted components such as gluten, dairy, nuts, and shellfish. This classification capability helps airlines produce accurate, comprehensive labels that can be printed on in-flight meal packaging (D. Wang et al., 2024) or made available through digital channels, such as seat-back screens or mobile apps. In addition to providing comprehensive ingredient lists, AI-driven labeling systems can also enhance the accuracy and consistency of allergen warnings. Mislabeling or omitting allergen information poses serious risks to passengers with food allergies, who may experience severe reactions if exposed to allergenic substances. AI systems help prevent these issues by cross-referencing ingredient data from suppliers and caterers (Muszyński et al., 2022) to ensure that all potential allergens are accurately listed. With advanced algorithms, AI can automatically flag any discrepancies between reported ingredients and known allergenic compounds, alerting food safety teams to potential risks (Jafrasteh et al., 2023). This level of detail in ingredient documentation minimizes human error in labeling, ensuring that passengers can trust the information provided about in-flight meals. AI also improves ingredient transparency by allowing airlines to provide passengers with information about the sourcing and production of food items (Baxter, 2020). Many passengers, especially those concerned with ethical and sustainable sourcing, appreciate knowing where and how their food was produced. AI can collate data from multiple suppliers and compile comprehensive reports on food origins (Vilas-Boas et al., 2023), sustainability practices, and quality assurance measures. This information can be displayed digitally, giving passengers insight into the provenance of their meals and increasing their trust in the airline's commitment to food safety and ethical sourcing. For airlines operating across international borders, where labeling standards vary significantly, AI systems can standardize ingredient and allergen information according to the strictest regulations. This global approach ensures that airlines remain compliant with the diverse regulatory requirements of different countries, reducing the risk of non-compliance and enhancing passenger safety (Hassan Metwaly et al., 2018). Furthermore, AI enables airlines to quickly adapt to changing regulations by automating updates to labeling requirements, thereby avoiding costly delays or disruptions in meal service (Rajaratnam, 2023). This adaptability is crucial for airlines, which operate under tight schedules and need efficient solutions to maintain food safety compliance without impacting service quality (Hassan et al., 2021).

6.2. Real-time alerts for passengers with dietary restrictions or allergies

Beyond static labeling, AI can deliver real-time, personalized alerts to passengers with specific dietary restrictions or allergies. For example, passengers who have indicated dietary needs, such as vegan, halal, kosher, or gluten-free options, can benefit from AI-driven customization (Mortensen Ernits et al., 2022) that allows them to view tailored information about available meals. When booking or during check-in, passengers can specify their dietary requirements, and AI algorithms can match them with suitable meal options based on their preferences. In-flight, passengers can use mobile apps or in-seat screens to access personalized meal recommendations, ensuring they make safe and informed choices. For passengers with allergies, the ability to receive real-time updates and alerts about in-flight meals is crucial to avoiding accidental allergen exposure (Klymenko, 2024). AI systems integrated with catering and inventory data can inform passengers about meal ingredients (Perera, 2024), highlighting potential allergens. For example, if a passenger with a nut allergy is served a meal with nuts, the system can alert both the passenger and flight crew to ensure safety precautions are taken. The system also updates dynamically based on the latest ingredient data, especially when substitutions are made due to supply chain disruptions or changes in the meal offerings. Additionally, AI-driven real-time alerts can notify the crew and passengers of food safety risks (Wong & Man, 2023). This transparency allows passengers to make informed decisions, reducing the risk of consuming meals exposed to unsafe conditions. AI-powered dietary alerts are further enhanced through mobile apps (Gami et al., 2024) that integrate with the airline's in-flight systems, allowing passengers to view ingredient details, allergen info, and meal status in real time. These apps also let passengers pre-select meals based on dietary needs, improving safety and enhancing the overall travel experience.

The role of AI in enhancing consumer safety and transparency extends beyond the technical provision of ingredient and allergen information; it also builds trust between airlines and passengers by demonstrating a proactive commitment to food safety. Passengers are more likely to trust airlines that transparently share food safety practices (Lamb et al., 2021), especially when advanced technology is used to safeguard their health. For example, in the rare event of an in-flight allergic reaction, AI systems that document and share ingredient data can provide crucial information to medical personnel, helping them respond more effectively. Additionally, this accountability is particularly important in the airline industry, where passengers expect a high level of care and safety. Furthermore, AI can support post-flight transparency by providing passengers with post-flight reports (Ceken, 2024; Verma, 2024) or digital records of the meals served, including detailed ingredient lists and allergen information. This level of detail is especially useful for frequent flyers with strict dietary requirements who may wish to track their in-flight meals and ensure ongoing consistency in the quality and safety of their food. By giving passengers access to comprehensive food safety information, airlines foster trust, which can lead to increased passenger loyalty and brand preference in a competitive industry (Klophaus & Yu, 2023). Furthermore, AI's ability to enhance transparency and ensure regulatory compliance strengthens consumer trust, creating a safer, more reliable environment for airline passengers worldwide. As AI technology continues to evolve, it will likely play an even more integral role in addressing the complexities of food safety and transparency in the airline industry, setting new standards for passenger care and satisfaction (Ernits; Mortensen Ernits et al., 2022).

7. Challenges in implementing AI in airline food safety

Implementing AI in airline food safety holds immense potential, but it also presents a host of significant challenges that airlines must overcome to ensure its success and sustainability (Igwe et al., 2023; Paraschi et al., 2024). Given the unique operating conditions of airlines where stringent regulations, international standards, and passenger expectations must be met introducing advanced AI solutions into food safety requires careful consideration of data privacy and security (Omweno, 2022; Rajaratnam, 2023), substantial initial and operational costs, and the complexities of integrating AI into existing airline operations.

7.1. Data privacy and security concerns

A major challenge in deploying AI for airline food safety is ensuring data privacy and security (Omweno, 2022). AI solutions, particularly those reliant on the Internet of Things (IoT) and cloud-based analytics, necessitate constant data collection, processing, and sharing across multiple platforms and stakeholders. In the context of airline food safety, AI systems need access to sensitive information, such as passenger dietary preferences, health conditions, and even real-time location data, to effectively monitor and address food safety risks in a personalized and timely manner. This raises important privacy concerns, as handling such data without robust security measures could potentially lead to breaches of passenger privacy (Alabsi & Gill, 2021; Pillai &

Devrakhyani, 2020), exposing airlines to legal and regulatory consequences. Airlines must comply with strict data protection regulations such as the General Data Protection Regulation (GDPR) in Europe, which mandates stringent controls on how personal data is collected, stored, and used. AI systems must be designed to protect data privacy while simultaneously allowing airlines to make use of data insights to enhance food safety (Pillai & Devrakhyani, 2020). One way to address these concerns is through data anonymization, which removes personal identifiers before data analysis. However, ensuring that anonymization does not affect the accuracy and relevance of AI outputs is a challenge. Furthermore, continuous compliance with data protection regulations across multiple countries adds another layer of complexity, as each jurisdiction may have different rules governing data usage. Data security is another critical concern (Shiwakoti et al., 2022), especially since AI-driven systems for food safety require connectivity with external suppliers, onboard monitoring devices, and ground-based servers. Such connections introduce multiple points of vulnerability where hackers could potentially gain access to sensitive passenger information or operational data. Cyber-security threats, including data breaches, unauthorized access, and malware attacks, are growing concerns in today's highly connected world, and AI-based airline systems are no exception. Protecting data across these interconnected systems requires airlines to invest in sophisticated encryption (Kagalwalla & Churi, 2019), firewalls, and secure access protocols, all of which add to the complexity and cost of AI implementation. Moreover, in an era where airlines are increasingly relying on AI and digital solutions, data security is fundamental to maintaining passenger trust. Any breach or mishandling of personal data could damage an airline's reputation, causing passengers to lose confidence in the airline's commitment to their safety and privacy. As a result, airlines must adopt rigorous data governance strategies (Rosenow et al., 2020) that go beyond compliance, proactively ensuring data integrity, confidentiality, and availability across AI-powered food safety solutions.

7.2. Initial setup and operational costs

The high initial setup and operational costs associated with AI are formidable barriers for many airlines considering (Sun et al., 2021) AI-driven food safety solutions. Implementing AI involves a considerable financial investment not only in the AI technology itself but also in the necessary hardware, such as IoT sensors for monitoring food temperatures, humidity levels, and storage conditions in real time. These IoT sensors must be installed both on the ground, within catering facilities, and on board aircraft, where space is limited and environmental conditions vary. Such equipment needs to be robust and reliable under the unique conditions of air travel, withstanding temperature fluctuations, vibrations, and limited connectivity. Beyond hardware, AI requires powerful computing resources to process large datasets and generate actionable insights (Basharat, 2022; Gill et al., 2022), often necessitating high-performance servers or access to cloud computing infrastructure. Additionally, airlines must allocate funds for software development or customization, as pre-packaged AI solutions may not fully meet the unique requirements of the airline industry. Customizing AI software to handle specific food safety protocols, integrate with existing airline management systems, and comply with industry regulations requires substantial investment in specialized programming and system design. The operational costs of maintaining AI systems for food safety can be equally high. AI solutions require continuous monitoring, updates, and maintenance to ensure they remain effective and secure. Furthermore, as AI algorithms learn and adapt based on new data, they require periodic retraining and refinement to ensure their predictions and recommendations remain accurate over time (Wali & Bulla). For instance, algorithms used to predict food spoilage based on environmental data will need to be regularly updated with the latest climate data and adjusted for changes in supplier practices or ingredient storage conditions. The cost of such ongoing support, including the staffing needs for AI system monitoring and troubleshooting, can place a significant financial burden (Jawabreh et al., 2023) on airlines. Cost challenges are further compounded by the limited profit margins in the airline industry. While AI has the potential to reduce food safety incidents and minimize waste over the long term, airlines may be hesitant to invest heavily in AI technologies without a clear and immediate return on investment (ROI). The benefits of AI for food safety, such as improved passenger safety, regulatory compliance, and brand reputation (Falco et al., 2021; Ivanov et al., 2022), are often difficult to quantify in monetary terms, making it challenging for airlines to justify the upfront costs of AI implementation.

7.3. Challenges with integrating AI into existing airline operations

Integrating AI into the existing operations of an airline is a complex and time-intensive process (Abdulrahman et al., 2023; Alexopoulos et al., 2020; Foster & Rhoden, 2020). Airline operations are inherently complex (Chanthati, 2024), with multiple moving parts and a high degree of interdependence between different departments, such as catering, logistics, and in-flight services. Incorporating AI into these existing workflows requires careful planning to ensure that the technology aligns with existing processes and does not disrupt service continuity (Ronak, 2024). A key challenge in AI integration is compatibility with legacy systems (Adepoju et al., 2024), as many airlines still rely on outdated technology that can't easily integrate with modern AI applications. These systems may lack the data collection, sharing, and processing capabilities needed for real-time AI decision-making. Consequently, airlines may face costly and time-consuming upgrades or replacements before AI can be implemented, adding to the financial and operational strain (Kollia et al., 2021). Additionally, AI adoption requires training staff, including flight attendants, caterers, and food safety managers, to use and interpret AI-driven insights. This often involves changing established routines and adapting to new tools and procedures. The logistical challenges are especially significant in food safety, where AI must interact with multiple stakeholders across the supply chain, from food suppliers to in-flight personnel and airport authorities. For instance, AI systems monitoring food temperature (Kollia et al., 2021; Krishnan et al., 2024) from the moment ingredients are sourced to the time they are served onboard require close collaboration with food suppliers, who may use different data formats or tracking systems. Establishing standardized protocols for data collection and sharing between these entities is critical but challenging, as each entity may have its own policies and technical limitations. Moreover, the airline environment poses unique technical challenges for AI integration. In-flight AI operates with limited connectivity, hindering real-time data transmission and applications needing immediate feedback, like predictive maintenance for refrigeration or real-time food temperature monitoring. Reliable AI performance despite these constraints demands innovative solutions, such as offline storage or intermittent syncing, complicating design and raising costs (Hassoun et al., 2024). Another significant issue is ensuring that AI-generated insights and recommendations are actionable within the constraints of an airline's operational structure. While AI can detect patterns and predict food safety risks, acting on these insights in a timely and efficient manner requires coordination across departments. For instance, if an AI system flags a potential contamination issue with a specific batch of meals, ground staff, catering teams, and flight crews must respond rapidly to mitigate the risk. Developing a streamlined response protocol that allows different departments to communicate and act on AI-driven alerts is essential to maximize the value of AI, but it poses a considerable logistical challenge. Lastly, cultural resistance to AI is a non-trivial challenge within the airline industry, where many processes are still managed through human oversight and manual checks. Employees may be reluctant to embrace AI (Bihari Singh, Gaurav, Sarkar, Sharan Dangayach, & Lal Meena, 2024; Sebola et al., 2019), fearing that automation will replace their roles or introduce an impersonal approach to food safety. To

address these concerns, airlines must foster a culture that views AI as a tool to support, rather than replace, human decision-making. This requires clear communication about the goals and benefits of AI, along with reassurances that AI is intended to enhance food safety practices, not undermine the importance of human expertise.

Additionally, airlines operate across multiple jurisdictions, each with its own food safety regulations, which adds complexity to AI adoption. Navigating this regulatory fragmentation can hinder the integration of AI systems, as airlines must ensure compliance with diverse national and international food safety standards, labeling requirements, and auditing practices (Falco et al., 2021; W. Zhou, 2024). This makes it challenging to develop AI solutions that are flexible enough to meet varied legal demands across different regions. Furthermore, AI-driven food safety systems rely on consistent internet connectivity for real-time data transmission (Hassoun et al., 2024; Zainuddin et al., 2024), but during flights, connectivity is often limited (J. Zhang et al., 2019). This restriction can impede the effectiveness of AI applications that depend on continuous updates, making it essential to explore offline data processing or intermittent syncing solutions.

In conclusion, while AI holds great promise for enhancing food safety in the airline industry, its implementation presents numerous challenges. Ensuring data privacy and security, managing high setup and operational costs, and navigating the complexities of integration with existing systems are all formidable obstacles that airlines must address. To realize the benefits of AI for food safety, airlines must invest not only in technology but also in staff training, robust security measures, and cross-departmental coordination. Overcoming these challenges will require a strategic approach that balances technological advancement with the realities of airline operations, paving the way for AI to become a trusted and effective partner in ensuring food safety at 35,000 feet (Grader et al.) Some major challenges and their potential solutions are given in Fig. 3. However, SWOT Analysis is given in Table 3.

8. Case studies

8.1. Case studies of AI applications in airline food safety

Examining real-world case studies of AI applications in airline food safety sheds light on the practical challenges, innovations, and successes that airlines have experienced in implementing these technologies. While the airline industry as a whole has been relatively cautious in adopting AI-driven food safety measures due to the complexities and unique conditions of in-flight food service, certain pioneers have begun leveraging AI to streamline their food safety processes and improve passenger experience. Some examples are given in Fig. 4. One notable example comes from **Singapore Airlines**, which has integrated AI to enhance its in-flight food safety standards and efficiency in meal preparation (Abd Elmoaty Mohamed & Soliman, 2022; ElSoad et al., 2022; Soonthodu et al., 2022) and delivery. Partnering with food tech companies specializing in predictive analytics and IoT solutions, Singapore Airlines implemented AI-driven sensors throughout its catering facilities to monitor real-time temperature, humidity, and hygiene conditions. These sensors are integrated with machine learning algorithms that predict potential points of contamination based on environmental changes, thereby enabling ground staff to preemptively address issues before they escalate. The system also alerts staff when food needs to be transferred or reheated to maintain optimal freshness. The result has been a measurable decrease in food spoilage and waste, alongside increased compliance with stringent food safety standards. The key lesson from this case is the value of continuous monitoring and early risk

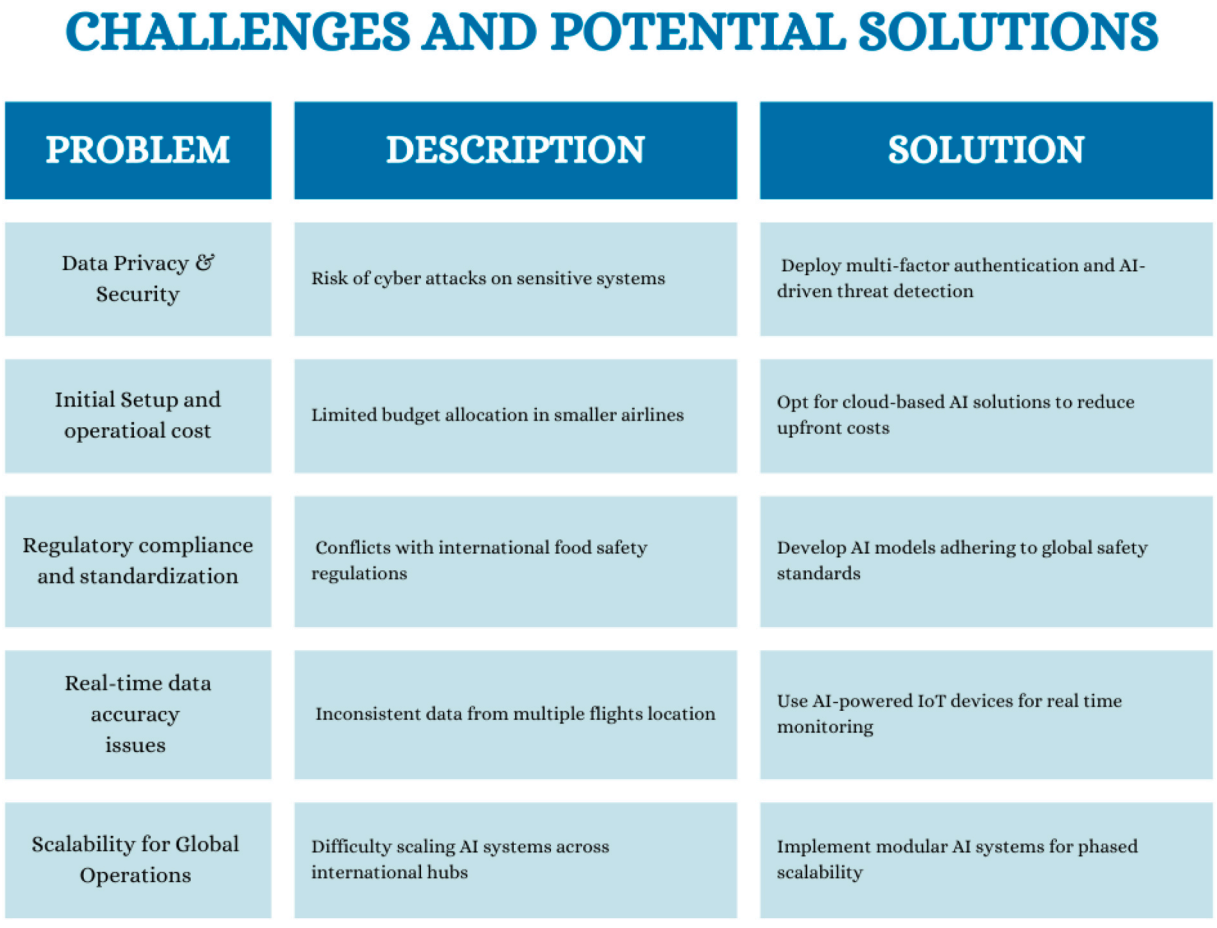


Fig. 3. Challenges and Potential solutions.

Table 3
SWOT analysis.

Strengths	Enhanced Monitoring & Detection	AI provides real-time monitoring for temperature, contamination, and spoilage, ensuring higher food quality standards (Alsaid et al., 2024).
	Increased Efficiency	Automated processes, such as quality control and inventory management, reduce labor costs and improve operational speed (Palakurti, 2022).
	Improved Compliance	AI ensures better regulatory compliance by automating documentation and record-keeping (Abid et al., 2024).
	Predictive Analytics	AI can forecast food demand and prevent waste, saving costs and reducing environmental impact.
	Traceability	Blockchain and IoT enhance the traceability of food items, making it easier to track and recall items in case of a contamination event.
Weaknesses	High Implementation Cost	Initial setup costs for AI technology, such as sensors and data infrastructure, can be high (Shaw et al., 2019).
	Dependence on Data Quality	AI requires large amounts of high-quality data to be effective, which can be difficult to manage across international supply chains.
	Complexity of Integration	Integrating AI with existing airline systems and food supply processes can be challenging.
	Technical Expertise Requirement	AI requires skilled personnel to manage, analyze, and interpret data, which may not be readily available in all airline settings (Kudashkina et al., 2022).
	Potential for System Errors	AI systems can malfunction or misinterpret data, which might lead to incorrect safety decisions if not monitored closely (Chanda & Banerjee, 2024).
Opportunities	Expansion of Predictive Capabilities	AI can be expanded to predict other safety risks, such as allergen presence or contamination trends (X. Wang et al., 2022).
	Enhanced Passenger Experience	AI-driven personalization can improve food safety and quality, contributing to better customer satisfaction.
	Partnerships with Tech Firms	Airlines can partner with AI tech companies for cutting-edge solutions tailored to their needs (Kumar, 2022).
	Reduced Environmental Impact	AI can help airlines reduce food waste through better demand forecasting and inventory management (Kanwal et al., 2024; L. Li, 2024).
	Adaptability to Emerging Regulations	AI can help airlines adapt to new food safety and traceability regulations efficiently.
Threats	Cyber-security Risks	Storing sensitive food safety and supply chain data digitally increases vulnerability to hacking or data breaches (Radanliev et al., 2024).
	Regulatory and Compliance Challenges	AI implementation may face regulatory challenges, especially in cross-border food safety standards.
	High Maintenance Costs	Continuous upgrades, monitoring, and maintenance of AI systems add to operational costs (Shaw et al., 2019)
	Reliability Concerns	Over-reliance on AI could lead to complacency in human oversight, potentially impacting safety if systems fail.
	Public Trust Issues	Passengers might have privacy concerns or trust issues if they perceive food safety to rely heavily on automated systems without human intervention.

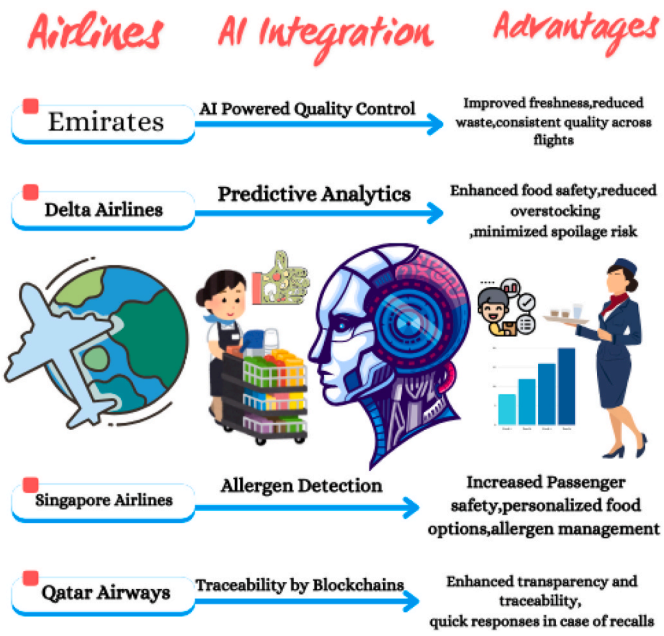


Fig. 4. Examples of AI integration for Food Safety in different Airlines.

detection, as it allows the airline to maintain food quality while reducing costs associated with food waste (Halizahari et al., 2021; Mortensen Ernits et al., 2022).

Another case involves **Delta Air Lines**, which deployed AI-powered predictive maintenance for its food storage (Rattanamanee, 2024) equipment as part of a larger initiative to improve food safety and operational efficiency. Delta implemented IoT sensors within refrigeration units used both on-ground and in-flight, which continually track temperature and equipment performance. By using predictive maintenance algorithms, the system identifies early signs of wear or potential failure, enabling maintenance teams to address issues before they compromise food safety. This has proven particularly beneficial in reducing incidents of equipment failure during flights, ensuring that food remains stored under safe conditions until it is served. This case highlights the importance of predictive maintenance in preventing food spoilage and contamination caused by faulty equipment, demonstrating that proactive AI applications can be an effective means of reducing food safety risks and improving service reliability (Y. Jiang et al., 2023; Ndhlovu et al., 2024; Yiğitol, 2025).

Japan Airlines (JAL) has also explored AI applications in food quality and supply chain management. By partnering with food safety and AI experts, JAL has introduced a blockchain-based traceability system powered by machine learning (Henderson, 2016; Kolasa-Sokołowska, 2021). This system enables the airline to track every ingredient used in their meals, from the original supplier through various stages of transportation, preparation, and delivery. Each stage of the food's journey is recorded in a secure digital ledger that cannot be altered, which ensures the authenticity and safety of ingredients and provides real-time transparency for any potential recalls or issues. The machine learning aspect of the system assesses historical data to forecast supply chain disruptions and make proactive adjustments to minimize delays and maintain ingredient quality (Chae et al., 2024). A significant takeaway from this case is that AI-powered traceability can be a critical tool in complex supply chains, particularly when airlines need to verify the quality and origin of perishable ingredients quickly.

In addition to these examples, several catering companies that supply major airlines, such as Gate Gourmet and LSG Sky Chefs (Redpath et al., 2017), have begun experimenting with AI to optimize food preparation and reduce waste. By using AI-driven analytics, these companies can forecast passenger demand for specific menu items on

flights, tailoring production volumes accordingly to minimize excess food production. This data-driven approach not only helps reduce waste (Kanwal et al., 2024) but also ensures that popular menu items are sufficiently stocked, improving passenger satisfaction (Lim & Lee, 2020). The primary lesson here is the importance of demand forecasting, as it aligns food production more closely with passenger preferences, benefiting both food safety and customer experience by reducing the risk of serving spoiled or low-quality food. These case studies underscore several key insights into AI's role in airline food safety. One of the main lessons is that proactive monitoring, through IoT and predictive analytics, is essential for maintaining quality under the time-sensitive conditions of in-flight service (Olufemi-Phillips et al., 2024). Furthermore, predictive maintenance has shown itself to be indispensable for managing the reliability of food storage devices, which directly impacts food safety by preventing failures that could lead to food spoilage.

Another lesson evident from these cases is that AI-powered traceability and supply chain transparency are highly effective for ensuring ingredient safety (Doherty et al., 2021), especially for airlines that source food from multiple international suppliers (Ma et al., 2022). Blockchain-backed AI solutions provide the added advantage of maintaining secure, immutable records, which bolster compliance with safety standards and allow for rapid responses in the event of recalls. By implementing AI in supply chain management, airlines can improve their resilience against disruptions while maintaining consistent food quality (Arvindaraj et al., 2023). Finally, these examples demonstrate that passenger satisfaction is directly linked to the airline's ability to provide fresh, high-quality meals. AI applications that incorporate demand forecasting contribute to both food safety and customer experience by ensuring that flights are stocked with the right amount of food, reducing spoilage while meeting passenger demand (Yusriza, 2022), (Yusriza, Abdul Rahman, Jraisat, & Upadhyay, 2023). In this way, AI enhances the operational efficiency of airline catering (Geske et al., 2024), enabling airlines to serve meals that meet both safety standards and passengers' expectations.

9. Future opportunities for AI in airline food safety

The future of AI in airline food safety holds immense potential (Bouzembrak et al., 2019; Yilmaz, 2024) as given in Fig. 5, with several emerging trends and technological advancements expected to further enhance food monitoring, compliance, and passenger safety in flight. As the airline industry faces evolving food safety standards and more complex regulatory requirements, AI-driven innovations stand to address these demands by enabling real-time monitoring, predictive analysis, and improved supply chain management tailored to airline operations. One of the most promising areas of AI development lies in advanced real-time food safety monitoring systems (Abass et al., 2024; Chhetri, 2024; Mu et al., 2024), which use a combination of IoT sensors, machine learning, and data analytics to continuously track temperature, humidity, and other critical variables throughout the lifecycle of airline meals. In the near future, AI-based sensors and smart packaging materials embedded with sensors could monitor these variables (Abekoon et al., 2024; Abimbola & Okpara, 2023; Musheer, Bharti, Mallira, & Yusuf), as well as detect microbial activity, in-flight atmospheric changes, and gas emissions from spoiled food. These sensors would provide real-time feedback, alerting flight crews and ground staff to potential risks before they escalate into safety hazards. For instance, smart packaging could use color-changing indicators to signal food freshness or spoilage, allowing crew members (Abdelhakim et al., 2019; Ghazal et al., 2019; Speakman & Grout, 2019) to verify food quality at a

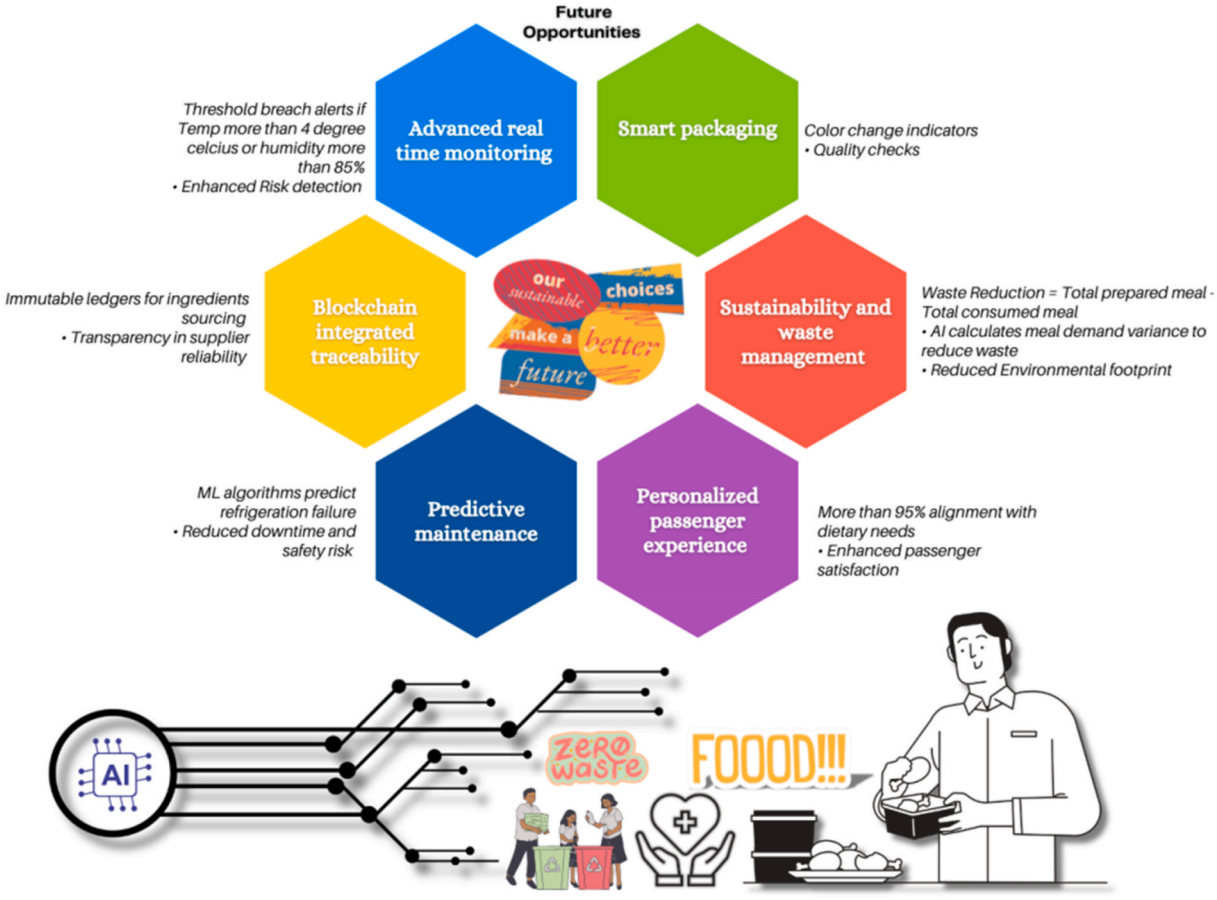


Fig. 5. Airline food safety future with AI technologies.

glance. These technologies, combined with machine learning algorithms that predict food safety risks based on historical data, would be especially valuable for long-haul flights, where food freshness is challenging to maintain over extended periods. The integration of blockchain technology with AI (Vyas et al., 2022; Q. Zhou et al., 2022) is also poised to transform the transparency and traceability of food supply chains for airlines. Blockchain, an immutable ledger that securely records transactions, can provide airlines with a complete and transparent record of each ingredient's journey from farm to in-flight meal tray. This record, when combined with AI's predictive analytics, could offer critical insights into supplier reliability and potential risks, such as contamination or fraud. In cases where specific ingredients must be traced back to their origins due to an outbreak or safety concern, blockchain-enabled AI systems could streamline the investigation process, helping airlines quickly identify and remove affected batches from circulation. Such systems would not only enhance food safety but also reinforce passenger trust by enabling airlines to demonstrate accountability and transparency in their sourcing practices. Predictive maintenance is another area where AI is expected to bring significant advancements (Salawudeen & Yusuff), particularly as it applies to the equipment used in airline food storage and preparation.

Machine learning algorithms could analyze performance data from refrigeration units, ovens, and other food-handling equipment to detect patterns indicating potential failures or inefficiencies (Hassoun et al., 2024). Airlines could use these insights to conduct targeted, timely maintenance, preventing equipment malfunctions that could compromise food safety. In the future, predictive maintenance systems may become even more sophisticated, automatically scheduling maintenance services and ordering replacement parts without human intervention. For instance, refrigeration units could adjust their internal conditions based on external temperatures and altitude, optimizing energy use and prolonging equipment lifespan while maintaining ideal food storage conditions. AI's role in evolving food safety standards is also set to expand, as regulators across the globe introduce stricter requirements for food handling, hygiene (Hwang et al., 2023; Park et al., 2020), and quality assurance in the airline industry. AI-driven solutions could help airlines stay ahead of these changing standards by streamlining compliance processes and automating record-keeping. For example, AI-powered compliance systems could monitor food safety records for accuracy and completeness, flagging any gaps that might expose the airline to regulatory penalties. Additionally, advanced AI systems could interpret and apply the latest regulatory guidelines, automatically updating processes to align with new standards, thereby reducing the administrative burden on airline staff (Ernits). This adaptability would be especially useful in a sector as globalized as the airline industry, where companies must navigate a complex web of national and international food safety regulations. Another promising trend in AI is the development of passenger-facing applications that offer increased transparency and personalization. For instance, airlines could leverage AI to create mobile apps or onboard screens displaying detailed information about in-flight meal ingredients, sourcing locations, and nutritional content. Passengers with dietary restrictions or food allergies (Ernits et al., 2022; Mortensen Ernits et al., 2022; Priya et al., 2020) could receive real-time alerts on which menu items meet their requirements, or even be offered customized meal options based on their dietary profiles. AI could analyze individual passenger data to suggest alternative menu items, recommend portion sizes, or propose meal options that align with specific health needs.

In the future, airlines might integrate these AI-driven personalization features with their loyalty programs, creating a highly individualized passenger experience while also reinforcing the brand's commitment to transparency and food safety. The use of AI to automate food preparation and assembly in catering facilities is another avenue with transformative potential. Robotics equipped with AI capabilities could handle tasks like portioning, packaging, and sealing meals, reducing human contact and thus minimizing contamination risks (Dalbhagat & Shukla). Robots could be programmed to follow stringent safety protocols and adjust their actions based on real-time data, ensuring consistent adherence to hygiene standards.

Additionally, future research needs to explore AI-powered robotics in airline catering for automated meal preparation, packaging, and quality control, enhancing efficiency and hygiene. Edge computing can enable real-time monitoring of food safety by processing data locally, reducing latency, and ensuring immediate responses to temperature or contamination risks. Advancements in predictive analytics and blockchain integration could further enhance traceability and compliance, making AI-driven food safety systems more resilient and adaptive to industry demands (Vasudevan, Lamine Mekhalfi et al., 2024; Vasudevan, Mekhalfi, et al., 2024). However, scaling these advanced AI technologies presents challenges, such as high initial costs, the need for robust infrastructure, and integration with existing airline systems. For instance, the implementation of AI-powered predictive maintenance and smart packaging may require significant investments in upgrading hardware and training staff (Sun et al., 2021), which could be a barrier for smaller airlines.

In advanced setups, AI-enabled robotics could even tailor meals based on flight data, adjusting portions based on route length, cabin class, and passenger demographics. This automation not only enhances food safety but also improves efficiency and consistency in meal production, making it easier for airlines to scale up or down based on demand fluctuations. AI's future in airline food safety also includes advancements in sustainability. With increasing pressure on airlines to reduce waste, AI could play a critical role in developing smarter waste management systems, such as monitoring food consumption patterns and recommending adjustments to reduce excess. Predictive analytics could help airlines optimize portion sizes, reducing food waste without compromising passenger satisfaction. AI algorithms could also suggest sustainable packaging alternatives (Lodhi et al., 2024; Spirito, 2024), calculate the environmental impact of menu items, and provide airlines with strategies to minimize their carbon footprint in meal production and service. Finally, AI is anticipated to support the development of rapid-response protocols for foodborne illness outbreaks and emergencies (L. Zhang et al., 2024). By analyzing real-time and historical data from multiple sources, AI systems could identify potential sources of contamination and recommend immediate remedial actions. These systems could also streamline communication between airlines, suppliers, and regulatory bodies, ensuring a coordinated response to food safety incidents. In the future, airlines could use AI-driven early warning systems to send real-time alerts when food safety risks reach critical levels (Dhal & Kar, 2025), enabling crew and ground staff to intervene before passengers are affected.

In summary, the future opportunities for AI in airline food safety are vast, driven by advancements in real-time monitoring, traceability, predictive maintenance, regulatory compliance, passenger personalization, automation, sustainability, and rapid-response protocols. Additionally, AI ethics in airline food safety must address bias in AI decision-making (Bondi-Kelly et al., 2023), ensuring fair risk assessments across diverse passenger needs. Passenger privacy concerns arise from collecting dietary and health data, requiring strict compliance with regulations like GDPR (Olsen & Wiesener, 2021). Responsible AI governance is essential for transparency, accountability, and ethical use of AI in food handling to maintain trust and safety. These innovations will enable airlines to elevate their food safety standards (Abd Elmoaty Mohamed & Soliman, 2022; Garg & Sharma, 2024; Rajaratnam, 2023), improve passenger trust and satisfaction, and meet the complex demands of a globalized, highly regulated industry.

10. Conclusion

In conclusion, the integration of Artificial Intelligence (AI) into the airline food safety sector offers transformative potential, addressing both current challenges and future demands. Throughout this paper, the

key advantages of AI have been explored, including enhanced real-time monitoring, predictive maintenance, improved supply chain management, and robust compliance with international food safety standards. AI has already proven its value in food supply chains by significantly reducing waste and improving efficiency. For example, AI-powered predictive analytics have been shown to reduce food waste by up to 30 % in some industries by optimizing procurement and forecasting inventory needs based on real-time demand data. This capability is particularly critical for the airline industry, where food spoilage and overstocking are common issues. AI innovations provide practical solutions to critical challenges faced by the airline industry, such as maintaining food quality and safety over long distances, ensuring transparency in sourcing, and mitigating the risks of foodborne illnesses during air travel. For instance, AI-driven IoT sensors can monitor and control temperature fluctuations during transport, a factor that causes up to 20 % of food losses globally. By addressing these issues proactively, airlines can not only improve food safety but also save millions of dollars annually that would otherwise be lost to spoilage and recalls. Traditional approaches to food safety in airlines are often reactive and inefficient, lacking the real-time responses or traceability that AI-powered systems provide. Machine learning, IoT-enabled sensors, and blockchain technologies enable airlines to trace each ingredient's journey, ensuring compliance with international food safety standards. This level of transparency has proven to increase consumer trust, with studies showing that 73 % of consumers prioritize food traceability when making purchasing decisions.

However, this paper emphasizes the long-term sustainability and scalability of AI-driven solutions for food safety in airlines. As regulatory requirements grow more stringent and passenger expectations for transparency and safety increase, AI's ability to analyze vast datasets and predict trends positions it as an indispensable tool in managing the complexities of the airline food supply chain. Moreover, implementing AI can lead to a 15–20 % reduction in operational costs by automating compliance verification, optimizing supply chain logistics, and minimizing manual interventions. The challenges of integrating AI into existing airline operations, such as data privacy concerns, high initial costs, and system compatibility, are acknowledged. However, with strategic planning, phased implementation, and cross-industry collaboration, airlines can overcome these barriers and achieve seamless AI integration. By doing so, they not only enhance food safety but also contribute to global sustainability efforts by reducing food waste and improving resource efficiency. Next steps for research should focus on developing industry-wide AI standards and ensuring regulatory alignment to facilitate seamless adoption. Collaboration between airlines, technology developers, and regulators will be crucial in overcoming data privacy concerns and infrastructure challenges. Continuous monitoring and iterative improvements of AI systems will also be essential to ensure adaptability as technology evolves, reinforcing long-term sustainability in the airline food safety sector.

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Data availability

Data will be made available on request.

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